You asked about $V = IR$;

ex. $V = IR$

$10V = 2A \times 5\Omega$

if we decrease one of the components

$10V = \downarrow 1A \times 10\Omega \uparrow$

as one component decreases, the other must increase.

Conservation of energy!

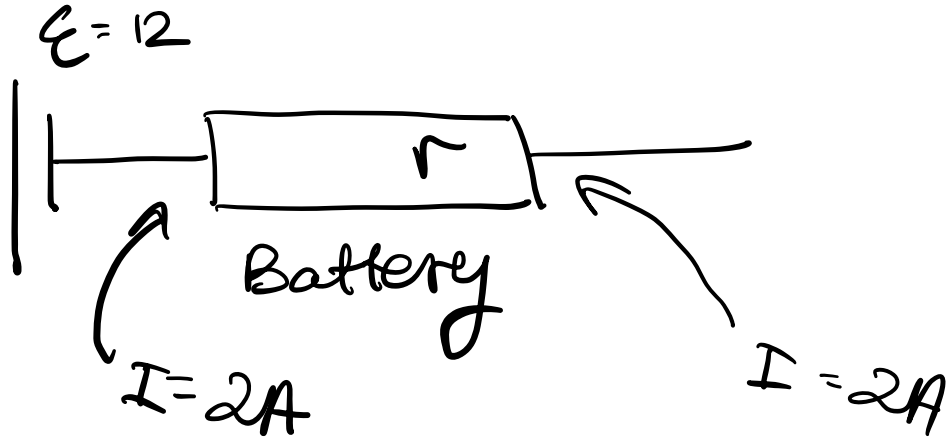
$\mathcal{E}$mf = electromotive force

Summary questions: p.222

1. a) Total $R_T = R + r =$
$= 1.5\,\Omega + 4.5\,\Omega = 6\,\Omega$

b) $I_T = \frac{V_T}{R_T} = \frac{12\,V}{6\,\Omega} = 2\,A$

c)



So lost pd: $V = Ir = 2 \times 1.5\,\Omega$
$= 3\,\Omega$

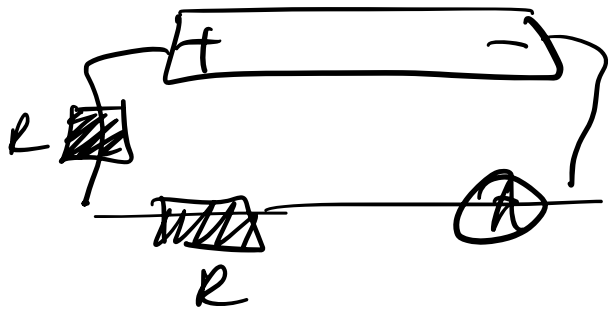
d) $12\,V - 3\,V = 9\,V$
because it does a "loop" around the circuit

2. a) $\mathcal{E} = IR + Ir =$

$= I(R+r)$

therefor $\therefore \quad I = \frac{\mathcal{E}}{R+r} = \frac{1.5V}{0.5+2.5} = 0.5A$

b) terminal so between + and − in battery



$V = \mathcal{E} - Ir = 1.5 - 0.5 \times 0.5 =$

$= 1.25\,V$

c) $P = VI = 1.25 \times 0.5 =$

$= 0.63\,W$

d) $P_{before} - P_{after} = \mathcal{E}I - VI =$

$= (1.5 \times 0.5) - (1.25 \times 0.5) =$

$= 0.75 - 0.625 = 0.13\,W$